

RED PINES, NB, Canada

WMO: 716340

Lat: 47.4394N

Lon: 65.5978W

Elev: 124

StdP: 99.84

Time Zone: -4.00 (NAA)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.4	-28.6	0.3	-23.2	-26.3	0.3	-21.3	6.1	-4.3	5.4	-6.8	1.1	0	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	29.4	19.9	27.4	19.1	25.7	18.2	21.5	26.2	20.6	25.0	19.7	23.9	2.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	14.9	23.2	19.1	14.0	22.5	18.2	13.3	21.8	63.2	26.4	59.8	25.0	56.9	23.8	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.6	4.2	-30.9	32.3	3.6	1.6	-33.5	33.5	-35.6	34.4	-37.6	35.3	-40.3	36.5		
(5)			-30.4	23.3	3.8	0.8	-33.2	23.8	-35.4	24.2	-37.6	24.7	-40.4	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-10.6	-9.8	-4.9	2.3	9.5	14.2	18.9	18.1	13.2	6.9	0.1	-6.3	(6)
(7)		DBStd	11.25	6.25	5.91	5.80	4.14	4.08	3.60	3.26	2.73	4.33	4.17	4.91	5.99	(7)
(8)		HDD10.0	2887	638	556	462	233	59	6	0	0	18	115	297	504	(8)
(9)		HDD18.3	5186	897	789	719	480	275	132	32	40	159	354	546	763	(9)
(10)		CDD10.0	841	0	0	1	3	44	132	276	250	115	19	1	0	(10)
(11)		CDD18.3	98	0	0	0	0	2	8	50	31	6	1	0	0	(11)
(12)		CDH23.3	1010	0	0	1	1	48	129	487	257	84	3	0	0	(12)
(13)		CDH26.7	240	0	0	0	0	9	33	132	46	19	0	0	0	(13)
(14)	Wind	WSAvg	1.7	2.0	2.0	2.2	2.1	1.8	1.5	1.4	1.3	1.4	1.6	1.8	1.8	(14)
(15)	Precipitation	PrecAvg	1038	85	68	81	72	92	84	95	88	82	104	92	94	(15)
(16)		PrecMax	1342	120	129	178	136	180	142	167	166	192	187	167	213	(16)
(17)		PrecMin	747	23	8	25	34	22	29	22	25	30	25	36	28	(17)
(18)		PrecStd	129	28	29	34	29	36	26	33	37	40	42	31	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	7.8	17.7	19.7	27.6	30.4	32.1	29.9	29.1	22.6	16.4	9.7	(19)
(20)			MCWB	9.3	5.8	9.7	10.3	17.5	18.5	21.1	20.6	19.9	16.4	15.7	9.0	(20)
(21)		2%	DB	4.1	3.8	8.7	14.9	24.3	26.8	30.0	28.0	25.9	18.8	12.5	6.5	(21)
(22)			MCWB	3.5	1.3	5.2	10.0	15.0	17.7	20.8	19.4	18.7	15.0	10.8	5.7	(22)
(23)		5%	DB	1.3	1.2	5.7	12.1	21.0	24.3	28.0	26.1	22.8	16.4	9.8	4.1	(23)
(24)			MCWB	0.5	-0.6	2.2	7.8	12.9	15.7	19.5	19.0	17.6	13.2	7.6	3.3	(24)
(25)		10%	DB	-0.7	-0.8	3.4	9.3	18.0	21.9	26.1	24.5	21.0	14.1	7.2	1.4	(25)
(26)			MCWB	-1.4	-2.1	0.5	5.1	11.8	14.9	18.7	18.3	16.8	11.3	5.6	0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	5.9	11.3	13.6	19.0	20.3	23.2	22.2	21.4	17.7	15.7	9.4	(27)
(28)			MCDWB	9.4	7.7	16.2	16.1	24.9	25.6	27.6	26.3	26.6	19.8	16.2	9.6	(28)
(29)		2%	WB	3.4	1.9	6.2	10.9	17.0	19.0	21.7	21.2	19.8	16.1	11.7	6.0	(29)
(30)			MCDWB	4.1	3.1	7.2	13.8	21.8	23.9	26.6	25.7	23.1	18.3	12.4	6.6	(30)
(31)		5%	WB	0.6	-0.4	2.8	8.0	14.8	17.6	20.8	20.2	18.8	13.9	7.9	3.8	(31)
(32)			MCDWB	1.1	0.8	4.8	12.2	18.2	22.3	25.4	24.3	22.1	16.6	9.1	4.4	(32)
(33)		10%	WB	-1.5	-1.9	0.7	5.7	12.9	16.3	20.0	19.3	17.4	11.8	5.6	1.0	(33)
(34)			MCDWB	-0.8	-1.0	2.5	8.8	17.5	20.0	24.7	22.9	20.7	13.7	7.1	1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	11.6	11.4	10.2	12.7	11.9	12.2	11.7	11.8	9.9	7.8	8.1	(35)
(36)			MCDBR	11.2	13.2	13.5	15.2	18.2	17.0	14.6	13.8	14.4	13.5	11.2	8.6	(36)
(37)			MCWBR	11.1	11.4	9.6	9.7	9.8	8.6	6.7	6.9	8.4	9.5	9.8	8.4	(37)
(38)		5% WB	MCDBR	11.5	11.7	11.2	13.5	13.9	12.6	11.9	11.4	12.4	11.2	10.1	8.2	(38)
(39)			MCWBR	11.6	10.6	9.0	9.9	8.7	7.4	6.3	6.4	8.5	8.8	10.0	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.276	0.311	0.364	0.363	0.382	0.412	0.387	0.355	0.326	0.295	0.277	(40)	
(41)		taud	2.378	2.348	2.362	2.329	2.404	2.349	2.302	2.368	2.445	2.519	2.548	2.462	(41)	
(42)		Ebn at Noon	814	897	912	889	901	881	848	856	854	831	796	776	(42)	
(43)		Edh at Noon	69	88	102	117	113	120	125	112	94	75	60	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.28	2.13	3.35	4.31	5.13	5.54	5.53	4.84	3.76	2.34	1.27	0.94	(44)	
(45)		RadStd	0.15	0.24	0.24	0.36	0.49	0.42	0.41	0.31	0.32	0.20	0.16	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page